A PROJECT REPORT

ON

ARDUINO BASED NOXIOUS GAS ALERTING SYSTEM WITH AMBIENT OBSERVATION FOR MINES

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE AWARD

OF

DIPLOMA IN EMBEDDED SYSTEMS ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled **"ARDUINO BASED NOXIOUS GAS ALERTING SYSTEM WITH AMBIENT OBSERVATION FOR MINES"** was successfully completed by **AVUTHU ROOPESH REDDY (21054-ES-025)** and his team members of fifth semester Diploma in Embedded System Engineering.

In partial fulfilment of the requirement for the award of Diploma in Embedded System Engineering GIOE (TS-SBTET, HYD) during the Academic Year 2023-24 as per curriculum C-21.

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ABSTRACT

This project presents the design and implementation of an Arduino based noxious gas alerting system with ambient observation for mines that combines the capabilities of the MQ-7 gas sensor and the DHT11 temperature and humidity sensor. The system aims to provide real-time monitoring of the air quality in the mines, specifically focusing on detecting carbon monoxide (CO) levels to ensure a healthy and safe living working environment and ambient temperature/humidity.

The MQ-7 gas sensor is employed to detect the presence of carbon monoxide in the environment. Carbon monoxide is a colorless, odorless gas that can be harmful when present in high concentrations. The gas sensor outputs an analog voltage proportional to the CO concentration, and the Arduino microcontroller processes this information to display the CO levels on the display.

In addition to gas detection, the DHT11 sensor is integrated into the system to monitor ambient temperature and humidity. The sensor provides accurate readings of these environmental parameters, which are also processed by the Arduino. Users can access real-time temperature and humidity information alongside the CO levels, offering a comprehensive overview of the environment.

The project incorporates a visual output system, such as an LCD display, to present the gathered data in a clear and user-friendly manner. Furthermore, the Arduino can be programmed to trigger alerts or notifications when the gas concentration exceeds predefined safety thresholds, enhancing the system's utility in maintaining a safe and comfortable environment.

This Arduino based noxious gas alerting system with ambient observation for mines serves as an accessible and cost-effective solution for individuals and organizations seeking to ensure a healthy and safe living working environment. The integration of the MQ-7 gas sensor and DHT11 temperature/humidity sensor provides a versatile platform for real-time monitoring and immediate response to potential environmental hazards.

TABLE OF CONTENTS

List of Tables

List of Figures

Chapter 1

Introduction

The main reason for death of miners is that due to any reason miners fall and losesconsciousness also proper treatment has not provided them at that time. According to a study, 418 people lost their lives and 438 people were injured in India during December 2020 to March 2022.

According to the National Institute for Occupational Safety and Health (NIOSH), one in 10 underground coal miners who have worked for at least 25 years have been identified as having black lung disease.

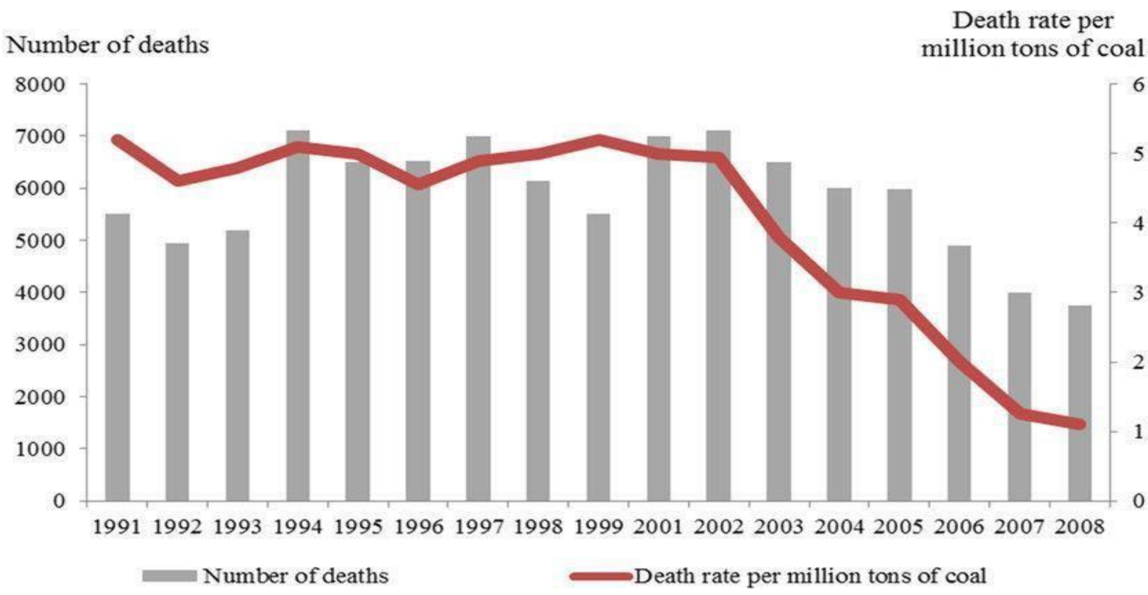

FIGURE 1

This system provides a sensor network for monitoring real time situation of underground mines from base station. It provides real time monitoring of harmful gases like CO, CH4, Temperature and Humidity.

When the carbon-monoxide gas concentration is beyond the critical level, controller triggers an alarm and LEDs for alerting the workers and keeps them safe by preventing an upcoming accident.

1.1 Objectives

The following are the objectives of this project:

- To provide a solution for miner’s safety.
- To detect the harmful gases around the miners and environmental data around them such as temperature and humidity and display to them.
- To alert the miners by alarm when the gases are detected.
- To make this project at low cost and affordable and make it such a way that requires less maintenance and easy to use.

1.2 Background and Literature Survey

In the present existing scenario, the miners are sent inside the coal mines without monitoring the environmental conditions inside mines, with the help of our project if there are any dangerous gases, the buzzer present in the device provides an alarm.

Chapter 2

Noxious gases alerting system with humidity and temperature working and real time management.

This chapter describes the proposed system, working methodology, software, and hardware details.

2.1 BLOCK DIAGRAM

The following block diagram (FIGURE 2) shows the system architecture of this project:

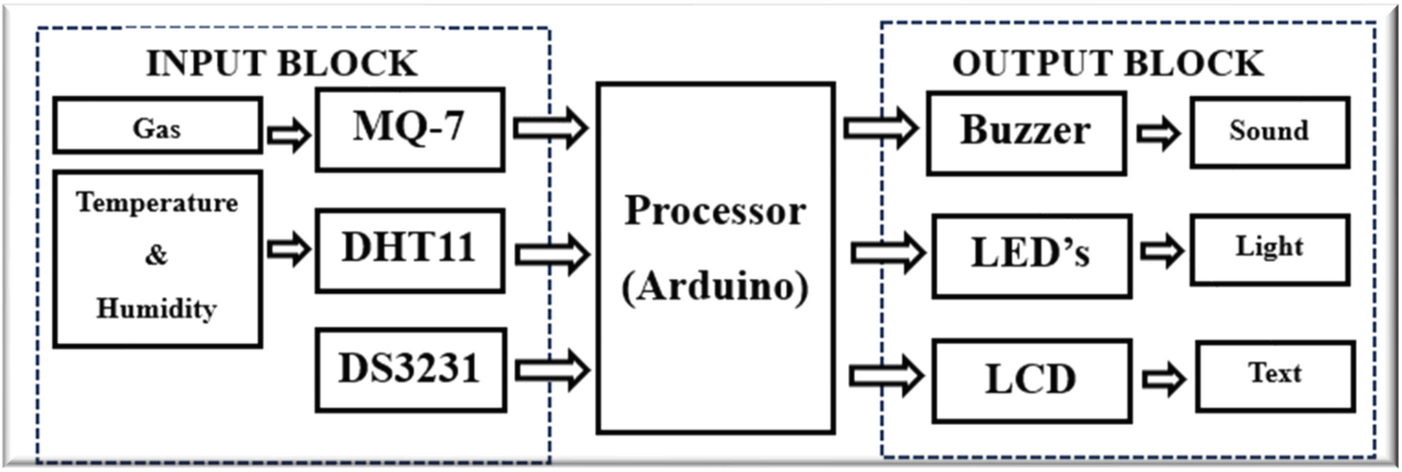

FIGURE 2

2.2 WORKING METHODOLOGY

The project is implemented to monitor various parameters in the coal mines such as leakage of gas, temperature, and humidity conditions. All the sensors are together considered as one unit and are placed in the coal mines. Here the gas is continuously monitored if any uncertainties in the level of gas arise, then buzzer and LEDs are used to alert the workers. Temperature and humidity values are also continuously monitored and displayed on the LCD display.

2.3 System details

This section describes the software and hardware details of the system.

2.3.1 Hardware Details

As shown in the figure 2 we have various hardware components being used in the system the details of each component are as follow:

- ARDUINO

Arduino is an open-source platform used for building electronics projects. It consists of both a physical programmable circuit board and a piece of software called the Arduino IDE (Integrated Development Environment) that runs on your computer. The Arduino IDE allows you to write and upload computer code to the physical board, enabling you to create interactive electronic objects.

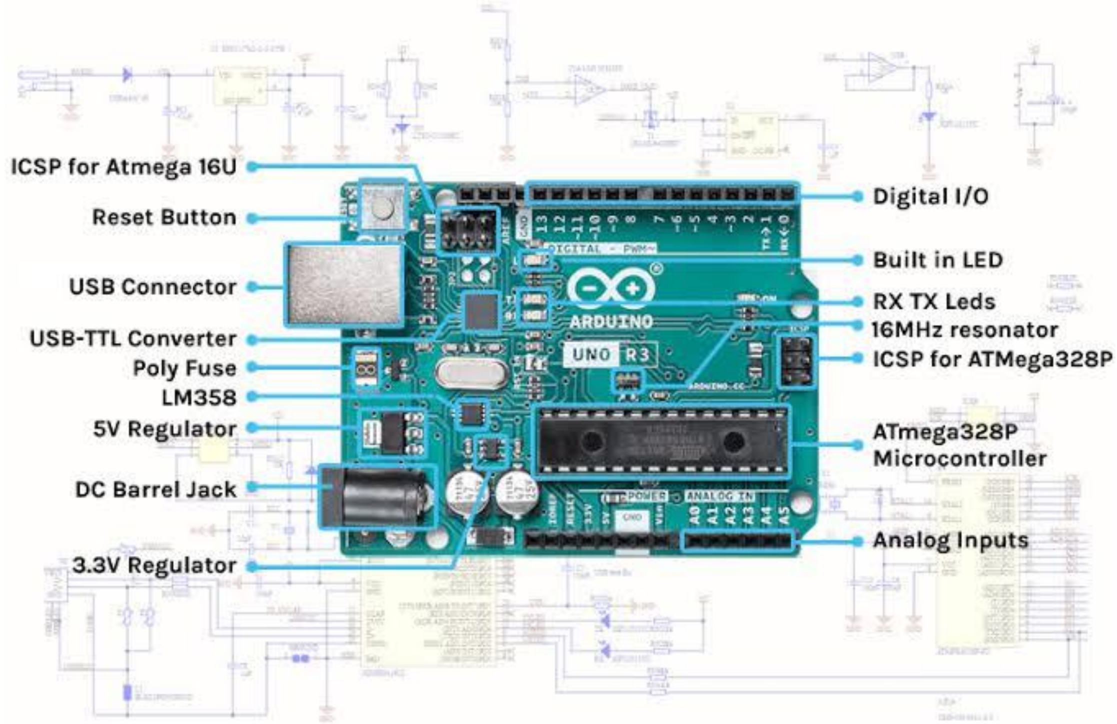

FIGURE 3

ARDUINO CONSISTS OF:

- ICSP for ATmega 16U
- Reset Button
- USB Connector
- USB-TTL Converter
- Poly Fuse
- LM358

- 5V Regulator
- DC Barrel Jack
- 3.3V Regulator
- Digital I/O
- Built in LED
- RX TX LEDs
- 16MHz resonator
- ICSP for ATMega328p
- ATmega328P Microcontroller
- Analog Inputs

- ICSP for Atmega 16U:

  ICSP pins are effective(helpful) to use when your USB port of Arduino is broken or damaged. These pins are used to code and boot an Arduino from an external source. These pins allow inter workings of two or more Arduino boards and also allow you to upload your firmware.

- Reset Button:

  The reset button is a hardware or software mechanism that allows you to restore a device or system to its original state or restart it in case of malfunction or freeze.

- USB Connector:

  The Arduino USB Cable for UNO and Mega (50 cm) is a type of USB cable designed specifically for the Arduino UNO and Mega microcontroller boards. It has a standard USB connector on one end and a Type B USB connector on the other, allowing it to be connected to a computer or other USB device.

- USB-TTL Converter:

  The USB to TTL Converter allows us to connect any serial port to USB Device, thereby enabling the mobility. This is used for USB to TTL Converter applications. Such as for data transmission, settings, debugging in various electronics devices.

- Poly Fuse

  Polyfuse is to protect the USB port, to keep your board from drawing too much power. The Uno has a polyfuse between the USB connector and the rest of the board.

- LM358

  This IC LM358 is 8-pin IC. This IC consists of two independent high-gain frequency compensated operational amplifiers designed to operate from a single supply or split supply over a wide range of voltages.

- DC Barrel Jack

  Barrel plug connectors are commonly used to interface the secondary side of a power supply with the device.

- Digital I/O

  The pins 0 to 13 are used as a digital input or output for the Arduino board.

- RX & TX LED’s

  RX and TX LED's blinks on Arduino board when successful communication takes place. By this two LED's we can understand communication is happening in between Arduino and serial device. TX LED blinks when Arduino is transmitting some data. RX LED blinks when Arduino receives some data.

- ATmega328P Microcontroller

  ATmega328P is a high-performance microcontroller, low power consuming 8-bit AVR microcontroller which is able to achieve the most single clock cycle execution of 131 powerful instructions as it uses advanced RISC architecture.

- Analog Inputs

  The analog inputs are used in control systems with input sensors that produce a voltage, current depending on the change in response to an environmental variation or system measurement.

- MQ-7 Gas sensor:

Gas detectors measure and indicate the concentration of certain gases in an air via different technologies. In the project we are using MQ-7 gas sensor, it's a sensor that detects an acute sensitivity to Carbon Monoxide and can detect the concentration of carbon monoxide in the surroundings. Sensor composed by micro AL2O3 ceramic tube, Tin Dioxide (SnO2) sensitive layer, measuring electrode and heater are fixed into a crust made by plastic and stainless-steel net. Sensitive layer of MQ-7 gas sensitive components is made of SnO2 with stability. So, it has excellent long-term stability. It's service life is 5 years under using condition.

FIGURE 4

- DHT-11 Temperature and Humidity sensor:

The Digital Humidity and Temperature – 11 (DHT-11) sensor features temperature & humidity complex with a calibrated digital signal output. By using the exclusive digital-signal-acquisition technique and temperature & humidity sensing technology, it ensures high reliability and excellent long-term stability. Each DHT11 element is strictly calibrated in the laboratory that is extremely accurate on humidity calibration. The calibration coefficients are stored as program in the OTP memory, which are used by the sensor's internal signal detecting process.

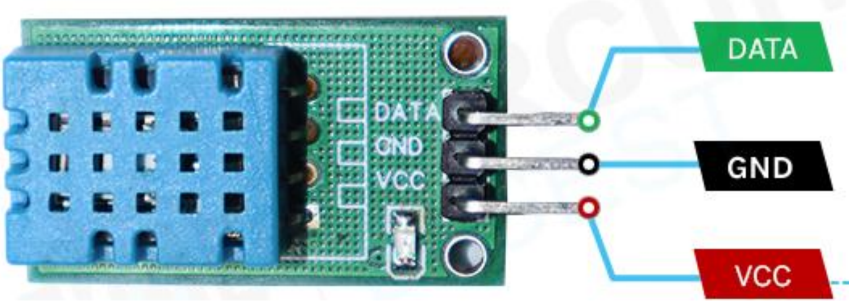

FIGURE 5

- DS3231 RTC:

The DS3231 is a low-cost, extremely accurate I2C real-time clock (RTC) with an integrated temperature compensated crystal oscillator and crystal. The device incorporates a battery input and maintains accurate timekeeping when main power to the device is interrupted. The integration of the crystal resonator enhances the long-term accuracy of the device. The RTC maintains seconds, minutes, hours, day, date, month, and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator.

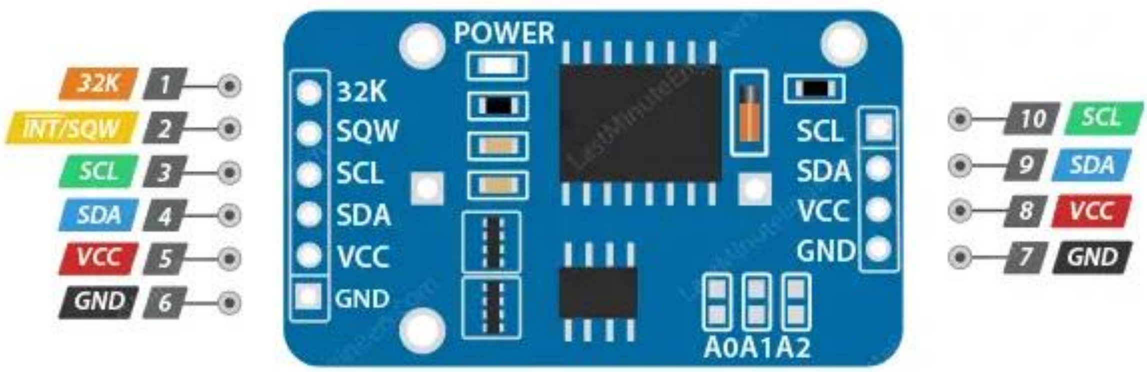

FIGURE 6

- LCD (16x2 Display):

To display interactive messages, we are using LCD Module. We used an LCD display of two lines, 16 characters per line that is interfaced to the controller. As LCD is very helpful in providing user interface as well as for debugging purpose. These LCDs are very simple to interface with the controller as well as are cost effective.

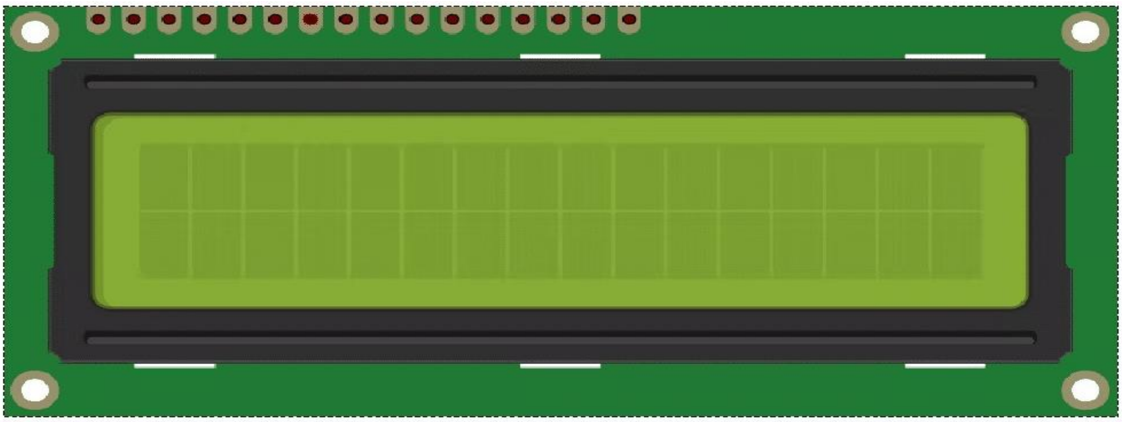

FIGURE 7

- BUZZER:
  A buzzer is an audio signalling device, which may be mechanical,electromechanical, or electronic. Here in the project, we are using active buzzer to raise an alarm and alert workers when the toxic gases are sensed around them.

FIGURE 8

- Breadboard
  A breadboard, solderless breadboard, or protoboard is a construction base used to build semi-permanent prototypes of electronic circuits. Unlike a perfboard or stripboard, breadboards do not require soldering or destruction of tracks and are hence reusable. A variety of electronic systems may be prototyped by using breadboards, from small analog and digital circuits to complete central processing units (CPUs).

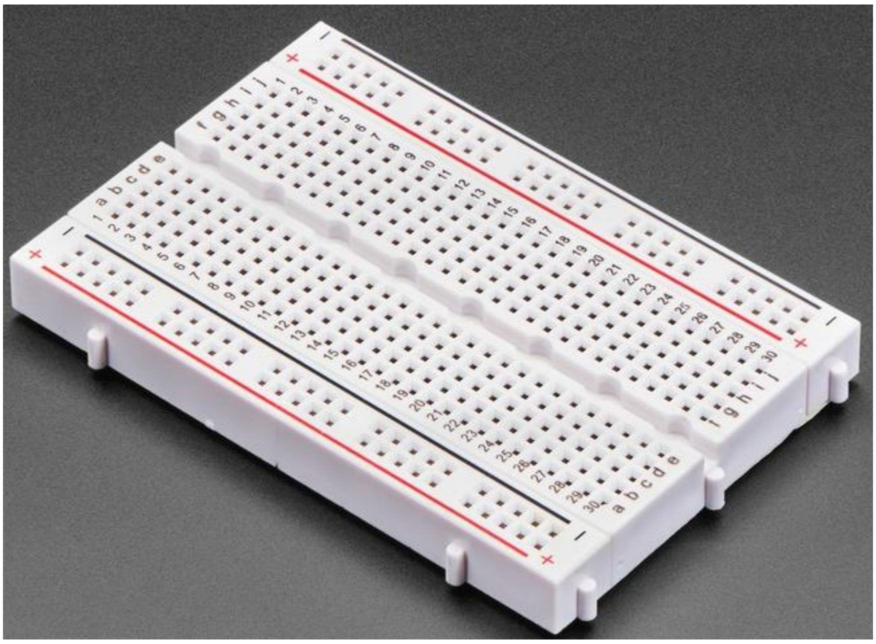

FIGURE 9

- Jumping wires / connecting wires:

  Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed.

FIGURE 10

- Battery
  An electric battery is a source of electric power consisting of one or more electrochemical with external connections for powering electrical devices.

FIGURE 11

2.3.2 SOFTWARE DETAILS

ARDUINO IDE:
Arduino is an open-source integrated development environment kit. We can download the Arduino IDE from the Arduino official website.

To download the Arduino, follow the following steps:

- Go to the official Arduino website.
- Click on downloads.
- Select the required version.
- Select the software which is compatible with your system (Windows, Mac, Linux, etc.) and download the software.
- Run the installer.
- Follow the steps.
- Install the required libraries.
- Click finish.

The Arduino Integrated Development Environment (IDE) serves as a versatile tool for programming Arduino boards.

Some of the uses of the Arduino IDE are the following:

1. Writing and Uploading Code:

    - The primary purpose of the Arduino IDE is to write, compile, and upload code to Arduino boards.
    - You can create programs (referred to as sketches) using the text editor within the IDE.
    - These sketches are saved with the file extension ".ino".

2. Offline Development:

    - The Arduino IDE allows you to work offline, making it ideal for users with poor or no internet connectivity.
    - You can write and upload code without relying on an internet connection.

3. Compatibility with Any Arduino Board:

    - The IDE is compatible with all Arduino boards.
    - Whether you're using an Uno, Nano, Mega, or any other model, the IDE provides a consistent development environment.

4. Two Versions Available:

    - There are currently two versions of the Arduino IDE:
        - IDE 1.x.x: The classic version that has been widely used.
        - **IDE 2.x:** A newer major release with enhanced features, including a modern editor, responsive interface, and advanced coding and debugging tools.

5. Basic Workflow:

    - To get started with the Arduino IDE:
        - Download and Install: Install either Arduino IDE 1.x.x or Arduino IDE 2.x based on your preference and platform.
        - Connect Your Board: Plug in your Arduino board to your computer.
        - Open the IDE: Launch the Arduino Software (IDE).
        - Write Code: Use the text editor to write your code.
        - Select Board and Port: Choose the appropriate board and port from the Tools menu.
        - Upload: Upload your sketch to the board.

Arduino IDE user interface:

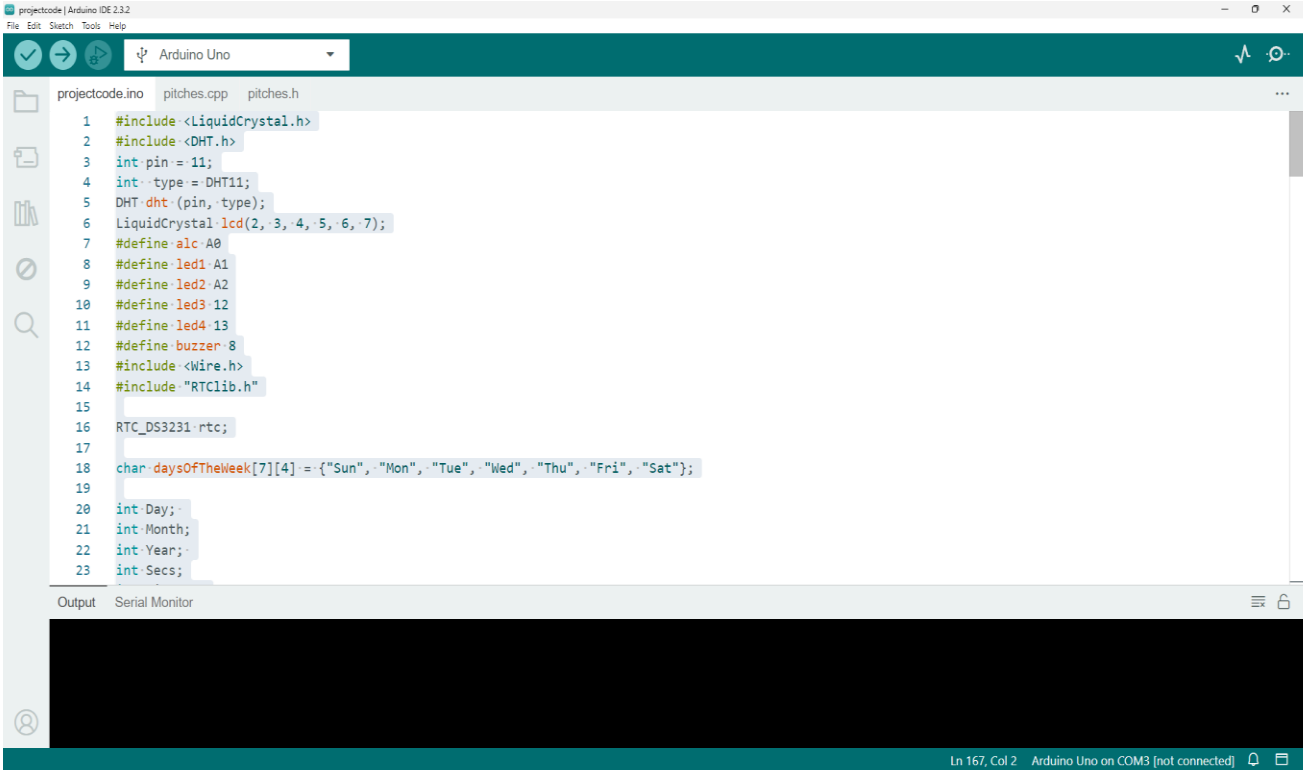

FIGURE 12

Source code:

```
#include <LiquidCrystal.h>
#include <DHT.h>
int pin = 11;
int  type = DHT11;
DHT dht (pin, type);

LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

#define alc A0
#define led1 A1
#define led2 A2
#define led3 12
#define led4 13
#define buzzer 8

#include <Wire.h>

#include "RTClib.h"

RTC_DS3231 rtc;

char daysOfTheWeek[7][4] = {"Sun", "Mon", "Tue", "Wed", "Thu", "Fri",
"Sat"};

int Day;
int Month;
int Year;
int Secs;
int Minutes;
int Hours;



String dofweek; // day of week

String myDate;

String myTime;
```

```
void setup()
{


Serial.begin(9600);
lcd.begin(16, 2);
dht.begin();
lcd.clear();

lcd.setCursor(0, 0);
 lcd.print("     WELCOME");
  delay(1000);
    lcd.clear();

lcd.setCursor(0, 0);
  lcd.print("  ARDUINO BASED");
     lcd.setCursor(0, 1);
      lcd.print("   NOXIOUS GAS");
       delay(2000);
        lcd.clear();

lcd.setCursor(0, 0);
  lcd.print(" ALERTING SYSYEM");
     lcd.setCursor(0, 1);
      lcd.print("  WITH AMBIENT");
       delay(2000);
        lcd.clear();

lcd.setCursor(0, 0);
  lcd.print("   OBSERVATION");
     lcd.setCursor(0, 1);
      lcd.print("    FOR MINES");
       delay(2000);
        lcd.clear();


pinMode(alc, INPUT);

pinMode(led1,OUTPUT);

pinMode(led2,OUTPUT);
```

```
pinMode(led3,OUTPUT);

pinMode(led4,OUTPUT);

pinMode(buzzer,OUTPUT);


if (! rtc.begin())
  {
    Serial.println("Couldn't find RTC");

    while (1);
  }

  rtc.adjust(DateTime(F(_DATE), F(TIME_)));

 digitalWrite(led4,HIGH);

 digitalWrite(led2,HIGH);


}
void loop()
{

float adcValue=0;
 adcValue = analogRead(alc);
   delay(10);
   float v= (adcValue/10) * (5.0/1024.0);
    float mgL= 0.67 * v;
     Serial.print("GAS:");
      Serial.print(mgL);
       Serial.print(" mg");
        lcd.setCursor(0,0);
         lcd.print("GAS: ");
          lcd.print(mgL,4);
           lcd.print(" PPM ");
            lcd.setCursor(0,1);
             if(mgL >= 20)
              digitalWrite(buzzer,HIGH);
               delay(500);
```

```
                digitalWrite(buzzer,LOW);
                 delay(5000);
                  lcd.clear();


float Temp = dht.readTemperature();
 lcd.setCursor(0,0);
  lcd.print("  Temperature:");
   lcd.setCursor(0,1);
    lcd.print("    ");
     lcd.print(Temp);
      lcd.print(" 'C");
       delay(5000);
        lcd.clear();

 if(Temp>=45)
        {
          digitalWrite(led3,HIGH);

          digitalWrite(led4,LOW);
        }

  else

        {
          digitalWrite(led4,HIGH);

          digitalWrite(led3,LOW);
        }

float Humid = dht.readHumidity();
 lcd.setCursor(0, 0);
  lcd.print("    Humidity:");
   lcd.setCursor(0, 1);
    lcd.print("     ");
     lcd.print(Humid);
      lcd.print("%");
       delay(5000);
        lcd.clear();
```

```
 if(Humid>=65)
        {

         digitalWrite(led1,HIGH);
         digitalWrite(led2,LOW);

        }

  else

        {

          digitalWrite(led2,HIGH);

          digitalWrite(led1,LOW);


        }

DateTime now = rtc.now();
lcd.clear();
Day = now.day();
Month = now.month();
Year = now.year();
Secs = now.second();
Hours = now.hour();
Minutes = now.minute();
dofweek = daysOfTheWeek[now.dayOfTheWeek()];

myDate = myDate +dofweek+ " "+ Day + "/" + Month + "/" + Year ;
myTime = myTime + Hours +":"+ Minutes +":" + Secs ;
// send to serial monitor
Serial.println(dofweek);
Serial.println(myDate);
Serial.println(myTime);

//Print on lcd

lcd.setCursor(0,0);
lcd.print(myDate);
```

```
lcd.setCursor(0,1);
lcd.print(myTime);
myDate = "";
myTime = "";
delay(3000);
lcd.clear();

}
```

CHAPTER 3

List Of Components and Cost Analysis

COMPONENT	COST
Arduino Uno	₹ 800
DHT11	₹ 85
MQ7	₹ 120
DS3231	₹ 130
16*2 LCD	₹ 100
Buzzer	₹ 10
Breadboard	₹ 50
Other Components	₹ 100
TOTAL	₹ 1400(apx)

TABLE 1

CHAPTER 4

APPLICATIONS

- Agricultural Industries
- Oil and Gas Industries
- Transportation Industries
- Indoor Air Quality Industries
- Medical and Life Sciences Field
- Food & Beverage

CHAPTER 5

RESULT

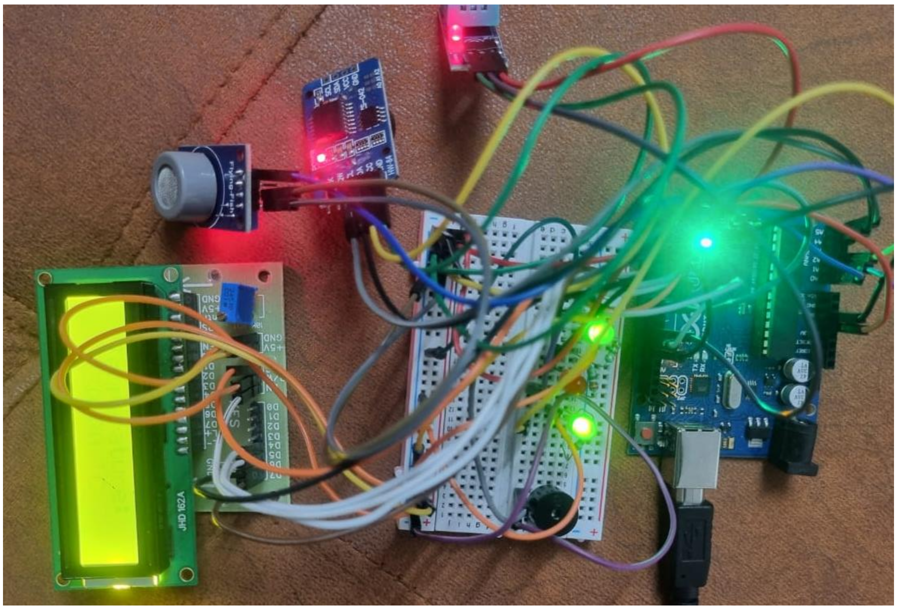

FIGURE 13.a

When we power up the project we can see the start up discription, after that we can see the readings of the gas sensor on the 16x2 LCD. We can see the gas reading in the loop. When the gas reading in the atmosphere change we can see the updated readings on the display. When the gas reading exceedes the set point value then the buzzer gets activated.

FIGURE 13.b

After the gas reading we can see the readings of the temperature sensor on the 16x2 LCD. We can see the temperature reading in the loop. When the temperature reading in the atmosphere change we can see the updated readings on the display. When the temperature reading is below the set point value then the green LED glows up,when it exceedes the set point value then the green LED turns off and the red LED lights up .

FIGURE 13.c

After the temperature reading we can see the readings of the Humidity sensor on the 16x2 LCD. We can see the humidity reading in the loop. When the humidity reading in the atmosphere change we can see the updated readings on the display. When the humidity reading is below the set point value then the green LED glows up,when it exceedes the set point value then the green LED turns off and the red LED lights up .

FIGURE 13.d

CHAPTER 5

CONCLUSION AND FUTURE WORK

We have implemented a project that comes from the field of embedded system. So that there is space for adding more devices and sensor that can be changed to the requirement of the user. The life of everyone is important as there is development in the machine and equipment that are used in a mine so there is safety of the workers. The above proposed system does not tackle all the problems but this help to monitor and collect the data in real time. This system is cost effective and has a lot of things that can be added like Wi-Fi module GPS module and etc. .

In case if any noxious occurs in the coal mine, then the system will alert the workers with the help of buzzer and LED's. This can help all workers who work in hazardous condition and can be upgraded as per their requirements.

REFERENCE

Arduino : https://docs.arduino.cc/hardware/uno-rev3/#featuresumentation

Software: https://www.arduino.cc/en/software

Circuit: https://www.circuito.io/app?components=512,11021

MQ-7 sensor interfacing: https://electropeak.com/learn/interfacing-mq-7-smoke-gas-sensor-module-with-arduino/

DHT-11 sensor interfacing: https://arduinogetstarted.com/tutorials/arduino-dht11

DS3231 RTC interfacing: https://lastminuteengineers.com/ds3231-rtc-arduino-tutorial/

LCD interfacing: https://circuitdigest.com/microcontroller-projects/interfacing-16x2-lcd-with-arduinouino (circuitdigest.com)

Reference Books:

1. Programming Arduino: Getting Started with Sketches by **Simon Monk**
2. Arduino Programming in 24 Hours by **Richard Blum**
3. Arduino: A Technical Reference by **J.M. Hughes**